IILM UNIVERSITY

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GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

Ethical Hacking

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION

June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Sanyog Raghav

Enrollment / Roll No.
2410030397 (Sec: 3CSE14)

Program & Degree: B.Tech (CSE) | Batch: 2024–2028

HOST ORGANIZATION

EduSkills Academy

Supported by AICTE National Internship Portal

Candidate & Program Overview

Candidate Details

Student Name: SANYOG RAGHAV

Roll Number: 2410030397

Section: 3CSE14

Degree & Branch: B.Tech (Computer Science & Engineering)

Academic Batch: 2026–2027

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: AICTE – EduSkills Virtual Internship

Specialization Domain: Ethical Hacking (english Language)

Duration: 8 Weeks (June 2026 – August 2026)

Performance Grade: Grade 'E' (Excellent: 80–89%)

Certificate ID: 47ae9c9fbc4550a778ce

Offer Letter Ref: C17Y2026M081462737

AICTE Student ID: STU6a2d013c18feb1781334332

Organization Profile: EduSkills Academy

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization driven by the mission "Nation Building Through Skills".

AICTE Collaboration: Operates under the patronage of the AICTE National Internship Portal (Ministry of Education, Govt. of India) to empower technical engineering talent across the nation.

Global Industry Alignment: Partners with global tech leaders including Cisco, AWS, Palo Alto Networks, Fortinet, Microchip, Red Hat, and Juniper Networks to bridge industry-academia readiness.

Core Focus Domains: Delivers high-impact hands-on learning in Cybersecurity, Ethical Hacking, Cloud Infrastructure, Artificial Intelligence, MLOps, and Enterprise Automation.

Problem Statement & Industry Context

⚠ The Industry Challenge

- Unprecedented rise in cyber threats including zero-day exploits, ransomware campaigns, APTs, and OWASP web vulnerabilities.
- Academic curricula often focus purely on theoretical cryptographic algorithms and textbook networking models.
- Engineering students lack hands-on exposure to how offensive attacks actually materialize in production environments.
- Theoretical knowledge alone leaves systems vulnerable without proactive adversary mindset and defensive validation.

The Solution Scope

- Complete end-to-end practical execution of the ethical hacking lifecycle: OSINT, scanning, assessment, exploitation, and reporting.
- Adversarial mindset training to discover, evaluate, and responsibly remediate vulnerabilities before malicious exploitation.
- Hands-on mastery of industry tools: Kali Linux, Nmap, Wireshark, Burp Suite, OpenVAS, Metasploit, and Snort IDS.
- Strict adherence to legal frameworks (IT Act) and safe simulation inside isolated virtual sandbox lab environments.

Core Project Objectives

Reconnaissance & OSINT

Perform passive & active footprinting, DNS interrogation, WHOIS lookups, and target attack surface mapping using tools like theHarvester, Shodan, and Maltego.

Scanning & Assessment

Execute network discovery, port scanning, and OS fingerprinting with Nmap/Wireshark; conduct automated vulnerability scans with OpenVAS and Nessus.

Web & System Hacking

Audit web applications against OWASP Top 10 flaws (SQLi, XSS, CSRF) via Burp Suite; simulate controlled system exploits and privilege escalation with Metasploit.

Defense & Governance

Implement defensive hardening, firewall configurations, Snort IDS rules, cryptographic protection (AES/RSA), and produce professional capstone audit reports.

Technologies & Security Toolchain

Lab Platforms & OS

Operating Systems: Kali Linux, Parrot Security OS, Ubuntu Server

Virtual Targets: Metasploitable 2/3, DVWA, WebGoat

Virtualization: VirtualBox, VMware Workstation

Recon & Network Analysis

Information Gathering: Whois, theHarvester, NSlookup, Dig, Shodan

Port & Service Scanning: Nmap (Network Mapper), Zenmap

Packet Sniffing: Wireshark, TCPdump, Netcat

Web Security & Exploitation

Vulnerability Scanners: OpenVAS, Nessus Community, Nikto

Web Proxy & Testing: Burp Suite, OWASP ZAP, SQLmap, Gobuster

Exploitation Framework: Metasploit (MSFConsole), Searchsploit

Auditing, Crypto & Defense

Password Auditing: John the Ripper, THC-Hydra, Hashcat

Cryptography: OpenSSL, GnuPG (GPG), HashCalc

Defense & Intrusion: Snort IDS, iptables firewall

Ethical Hacking & Pen-Testing Architecture

- **Reconnaissance Layer:** Passive OSINT & active DNS/domain footprinting to map target perimeter.
- **Scanning & Enumeration:** Nmap & Wireshark service detection, open ports, and live protocol analysis.
- **Vulnerability Assessment:** Automated Nessus/OpenVAS scanning & CVE matching against security flaws.
- **Web Application Security:** OWASP Top 10 auditing (SQLi, XSS, CSRF) with Burp Suite proxy interception.
- **Exploitation & System Hacking:** Targeted payload crafting & execution via Metasploit Framework.
- **Privilege Escalation Layer:** Evaluating misconfigurations, weak sudo rules, and local credential access.
- **Defense & Remediation:** Hardening with Snort IDS, iptables, patch management, and executive reporting.

PTES & Defense-in-Depth Pipeline

Phase 1: Target Recon & OSINT
theHarvester, Whois, Shodan, Dig

Phase 2: Network & Port Scanning
Nmap SYN/UDP Scan, Wireshark Inspection

Phase 3: Vulnerability Identification
Nessus, OpenVAS, Nikto CVE Matching

Phase 4: Web & System Exploitation
Burp Suite, OWASP ZAP, Metasploit MSF

Phase 5: Post-Exploit & Privilege Escalation
John the Ripper, Hydra, Linux Sudo Auditing

Phase 6: Defense Hardening & Reporting
Snort IDS Rules, iptables, Executive Report

Curriculum Timeline (Weeks 1 to 4)

Week 1
Ethical Hacking Fundamentals & Lab Setup

Cybersecurity concepts, CIA Triad, hacker classifications, ethical & legal compliance (IT Act), and isolated Kali Linux & VirtualBox setup.

Week 2
Reconnaissance, Footprinting & OSINT

Passive information gathering via search engines, WHOIS databases, DNS interrogation (dig, nslookup), email harvesting (theHarvester), attack surface mapping.

Week 3
Network Scanning & Protocol Analysis

Active host discovery, port scanning, TCP/UDP flag analysis with Nmap; live packet sniffing, protocol inspection, and MITM analysis using Wireshark.

Week 4
Vulnerability Assessment & Scanning

Vulnerability management lifecycle, running automated scans with OpenVAS/Nessus, analyzing CVE databases, assessing CVSS severity scoring.

Curriculum Timeline (Weeks 5 to 8)

Week 5
Web App Security & OWASP Top 10

Investigating web architecture flaws, auditing OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF, IDOR), HTTP proxy intercepting with Burp Suite and OWASP ZAP.

Week 6
System Hacking & Privilege Escalation

System penetration techniques, exploiting known services using Metasploit Framework, password cracking with John the Ripper/Hydra, Linux privilege escalation.

Week 7
Malware, Cryptography & Wireless

Understanding malware types (Trojans, ransomware), symmetric/asymmetric encryption, hashing algorithms (SHA, MD5), and Wi-Fi security (WPA/WPA2 handshakes).

Week 8
Applied Capstone & Assessment

Full-scale penetration testing simulation on vulnerable host instances, end-to-end vulnerability assessment report writing, mitigation roadmaps, and final exam.

Key Projects & Demonstrated Skills

- ✓ **Network Reconnaissance & Discovery:** Performed thorough passive OSINT reconnaissance and active network scans using Nmap, Wireshark, and DNS tools to map network topology and active services.
- ✓ **Automated & Manual Vulnerability Assessment:** Evaluated live sandbox environments using OpenVAS, Nessus Community, and Nikto, correlating CVE findings against CVSS impact scores.
- ✓ **OWASP Top 10 Web Application Testing:** Conducted in-depth security audits against web applications, successfully identifying and testing SQL Injection, Cross-Site Scripting (XSS), and Broken Access Controls using Burp Suite.
- ✓ **System Exploitation & Privilege Escalation:** Simulated controlled multi-stage exploits via Metasploit Framework (MSFConsole), demonstrated password auditing using John the Ripper/Hydra, and verified Linux privilege escalation vectors.
- ✓ **Defensive Hardening & Comprehensive Audit Reporting:** Configured Snort IDS detection rules, implemented iptables firewall policies, and produced an executive penetration testing report with prioritized mitigation recommendations.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	Ethical Hacking Virtual Internship (english Language)
Issuing Organization	EduSkills Academy (ISO 9001:2015 & ISO 20000-1:2018 Certified)
Academic Patronage	AICTE National Internship Portal & EduSkills Foundation
Certificate ID	47ae9c9fbc4550a778ce (Issued: July 2026)
Offer Letter Ref	C17Y2026M081462737
AICTE Student ID	STU6a2d013c18feb1781334332
Performance Grade	**Grade 'E' (Excellent: 80–89%)**

Certificate & Official Credential Preview

EduSkills Academy | AICTE Virtual Internship

CERTIFICATE OF VIRTUAL INTERNSHIP

This is to certify that

SANYOG RAGHAV

IILM University, Greater Noida
has successfully completed the 8-weeks virtual internship program in

Ethical Hacking (english Language)

During June – August 2026 | Awarded Grade 'E' (Excellent)

Certificate ID: 47ae9c9fbc4550a778ce • AICTE Student ID: STU6a2d013c18feb1781334332

Thank You!

Questions & Discussions

Student Presenter

Sanyog Raghav

Roll No: 2410030397
Section: 3CSE14
Batch: 2024-2028

Institution

School of Computer Science & Engineering

IILM University
Greater Noida, Uttar Pradesh